

Recap

- Propositions
- Logical connectors
 $\wedge, \vee, \neg (\neg), \oplus, \Rightarrow, \Leftrightarrow$
- Truth tables
- Tautology

ex: $p \vee \neg p$

Def : Inverse, converse and contrapositive statements of an implication are as follows:

Given two propositions p, q and the implication $p \Rightarrow q$

converse of the implication : $q \Rightarrow p$

Inverse of the implication : $\neg p \Rightarrow \neg q$

contrapositive statement of the implication : $\neg q \Rightarrow \neg p$

Ex: If It rains in Bozeman, then Bozeman is wet

$\underbrace{\text{If It rains in Bozeman}}_p, \text{ then } \underbrace{\text{Bozeman is wet}}_q$

$$p \Rightarrow q$$

converse of $p \Rightarrow q$: $q \Rightarrow p$

If Bozeman is wet then it rains in Bozeman

Inverse of $p \Rightarrow q$: $\neg p \Rightarrow \neg q$

If It does not rain in Bozeman then Bozeman is not wet.

Contrapositive statement of $p \Rightarrow q$: $\neg q \Rightarrow \neg p$

If Bozeman is not wet then it does not rain in Bozeman.

p	q	$p \Rightarrow q$	$\neg p$	$\neg q$	$q \Rightarrow p$	$\neg p \Rightarrow \neg q$	$\neg q \Rightarrow \neg p$
0	0	1	1	1	1	1	1
0	1	1	1	0	0	0	1
1	0	0	0	1	1	1	0
1	1	1	0	0	1	1	1

Logical Equivalence

Two propositions φ , ψ are logically equivalent, written as $\varphi \equiv \psi$, If they have exactly identical truth tables.

(In other words, their truth values are same under every truth assignment)

ex. consider $\neg(p \wedge q)$, $\neg p \vee \neg q$

p	q	$\neg p$	$\neg q$	$p \wedge q$	$\neg(p \wedge q)$	$\neg p \vee \neg q$
0	0	1	1	0	1	1
0	1	1	0	0	1	1
1	0	0	1	0	1	1
1	1	0	0	1	0	0

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\overline{p \wedge q} \equiv \sim(p \wedge q)$$

De Morgan's law for sets

$$\overline{A \cup B} = \bar{A} \cap \bar{B}$$

$$\overline{A \cap B} = \bar{A} \cup \bar{B}$$

for propositional logic

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$
$$\overline{(p \vee q)}$$

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$
$$\overline{(p \wedge q)}$$

Proof by contradiction

P

$\neg P$

Idea: Suppose we want to prove that Proposition ψ is true. Instead of proving ψ is true; we show that ψ cannot be false.

Def: proof by contradiction

To prove φ using proof by contradiction we assume the negation of φ and derive a contradiction; that is we assume $\neg\varphi$ and prove $\neg\varphi$ is false.

Theorem: Let $n \in \mathbb{Z}$, If $\overbrace{n^2 \text{ is even}}^P$, then $\underbrace{n \text{ is even}}_Q$.

$$P \Rightarrow Q \equiv \neg P \vee Q$$

$$\neg(P \Rightarrow Q) \equiv \neg(\neg P \vee Q) \equiv \neg(\neg P) \wedge \neg Q \equiv P \wedge \neg Q$$

$$n^2 \text{ is even and } n \text{ is odd} \equiv P \wedge \neg Q$$

Trying to prove $P \Rightarrow Q$ by proof by contradiction.

Assume the negation of the initial claim is true.

$$n \in \mathbb{Z} \wedge n^2 \text{ is even} \wedge n \text{ is odd}$$

by assumption

$$n = 2k + 1, k \in \mathbb{Z}$$

by def of odd

$$n^2 = (2k+1)^2 = 4k^2 + 4k + 1$$

by algebra.

$$n^2 = 2 \cdot (\underbrace{2k^2 + 2k}_c) + 1$$

by factoring

$$n^2 = 2 \cdot c + 1, \quad c \in \mathbb{Z}$$

sum of product of
ints

n^2 is odd

by def of odd

This is a contradiction, we assumed that n^2 is even and n is odd, but we got a contradiction that n^2 is odd.

Therefore, our initial assumption is incorrect.
Hence, the original claim is true $\square$.

Claim : $\sqrt{2}$ is not rational.

Try to prove ^P this using proof by contradiction.

Assume negation of the claim ($\neg P$)

$\sqrt{2}$ is a rational number by assumption.

$\sqrt{2} = \frac{p}{q}$, $p, q \in \mathbb{Z}$, $q \neq 0$ by def of rationals.

$\sqrt{2}^2 = \left(\frac{p}{q}\right)^2$ by algebra

$2 = \frac{p^2}{q^2}$ " "

$p^2 = 2q^2$ " "

p^2 is even by def of even

p is even by previous theorem.

$p = 2k$, $k \in \mathbb{Z}$ by def of even.

$(2k)^2 = 2 \cdot q^2$ by substitution.

$2k^2 = q^2$ by algebra

q^2 is even

by def of even

q is even

by previous theorem

p, q are even

p, q have a common factor 2.

because p, q are even.

But this is a contradiction because p, q cannot have common factors.

$\therefore$ our initial assumption is incorrect

Hence $\sqrt{2}$ is not rational.

$\square$.